MIL-P-63084 (PA)
23 February 1977

MILITARY SPECIFICATION

PROPELLANT, BALL POWDER

FOR

CARTRIDGE, 105MM, HE, XM710

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers propellant for use in ammunition (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-D-98	- Diphenylamine
MIL-N-244	- Nitrocellulose
MIL-N-246	- Nitroglycerin
MIL-A-48078	- Ammunition Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-286	- Propellant, Solid, Sampling, Examination and Testing
MIL-STD-652	- Propellants, Solid, For Cannons Requirements and Packing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1376

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite, "49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 Constituent Material - The constituent materials shall comply with the requirements of the following applicable specification:

Constituent Material	Conforming to Specification
Diphenylamine	MIL-D-98
Nitrocellulose	MIL-N-244 (Except Nitrogen Content 13.05 to 13.20%
Nitroglycerin	MIL-N-246 Type I

3.2 Composition and Physical Properties -

3.2.1 Composition - The propellant composition shall comply with the requirements given in Table I. The limits are not adjusted to a volatiles free basis. (See 6.2) when tested as specified in 4.5.1.

Table I

Constituents	Percent
Nitrocellulose, Minimum	60.0
Nitroglycerin	35.0 ± 2.00
Diphenylamine, Added	0.75 to 1.50
Total Volatiles, Max	2.0
Moisture & Volatiles	0.35 to 0.85

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Dibutylphthalate, Max.	1.00
Dinitrotoluene, Max.	1.00
Graphite, Max.	1.00

3.2.2 Physical Properties - The propellant shall comply with the requirements given in Table II when tested as specified in 4.5.1.

Table II

Test	Value
Heat Test - 120°C Methyl violet test paper shall not change to salmon pink color in less than 40 minutes. Explosion shall not occur in less than 5 hours.	
Dust and Foreign Matter, Max %	0.10
Bulk Density, gm/cc	0.630 to 0.730
Heat of Explosion, cal/gm Nominal	1190 to 1220
Diameter, inches, Nominal	.027/.020
Thickness, inches, Nominal	0.007

3.3 Closed Bomb - The propellant shall have a relative quickness of 100 ± 10 percent when tested as specified in 4.5.2.

3.4 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract (See 6.3).

3.5 Workmanship (See 6.2) - The best commercial practices shall be used in the formulation of propellant furnished under this specification and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of Inspections - The following types of inspection shall be conducted on this item:

a. First Article Inspection (See 4.3).

b. Quality Conformance Inspection (See 4.4).

c. Packing (See 5).

4.3 First Article Inspection -

4.3.1 Submission - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 5 pounds of propellant obtained by sampling as described in 4.4.2. The samples shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be Performed - The sample will be subjected by the government to any or all of the examinations or tests specified in 4.5 of this specification.

4.3.3 Rejection - See MIL-A-48078.

4.4 Quality Conformance Inspection -

4.4.1 Inspection Lot Formation - Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Sampling - The sampling shall be in accordance with MIL-STD-652, except that a five pound sample shall be randomly collected for chemical and physical tests. If the sample fails to meet any of the requirements of paragraph 3, the lot shall be rejected. One retest shall be permitted for the closed bomb test: No other retest are permitted.

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4.4.2.1 Classification of Defects - The classification for the test shall be as specified in Table III.

Table III

Test	Defects Classification
Nitrocellulose	Major
Nitroglycerin	Major
Diphenylamine	Major
Total Volatiles	Major
Moisture & Volatiles	Major
Bulk Density	Major
Dust and Foreign Material	Major
Heat Test	Major
Closed Bomb	Major
Diameter	Informational
Heat of Explosion	Informational
Thickness	Informational
Workmanship	Minor
Dibutylphathalate	Minor
Dinitrotoluene	Minor

4.4.3 Inspection Equipment (See 6.4) - The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment shall be employed where applicable for all tests and examinations specified in 4.4 and 4.5. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purpose.

4.5 Test Methods and Procedures - The tests in 4.5.1 through 4.5.2 shall be performed using prescribed analytical procedures for duplicate determinations given in standard analytical text-books (See 6.5).

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4.5.1 Chemical and Physical Properties - The chemical and physical properties shall be determined utilizing the MIL-STD-286 method shown in Table IV.

Table IV

Properties	Methods
Nitrocellulose	209.2
Nitroglycerin	208.2 or T208.5
Diphenylamine	217.3 or T208.5
Dibutylphthalate	204.1 or T208.5
Dinitrotoluene	205.2 or T208.5
Graphite	309.1
Total Volatiles	103.1 or T103.5
Moisture & Volatiles	101.2
Heat Test	404.1
Dust and Foreign Matter	501.1
Bulk Density	502.1
Heat of Explosion	802.1
Diameter	506.1
Thickness	504.1

4.5.2 Closed Bomb Test - The closed bomb test shall be conducted in accordance with method 801.1 of MIL-STD-286. A 200 cc nominal closed bomb at a loading density of 0.1 gm/cc shall be used. (WC 230 Lot SM5 as standard).

5. PREPARATION FOR DELIVERY

5.1 Packing Level A - The packing shall be in accordance with MIL-STD-652.

5.2 Alternate Packing (Level C - Commercial Pack) - Unless otherwise specified by the Contracting Officer, the propellant shall be packed in standard commercial containers acceptable by common or other carrier for safe transportation to the point of delivery.

6. NOTES

6.1 Intended Use - The propellant of this specification is intended for use as a propelling charge in the Cartridge, 105MM, HE, XM710.

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6.2 Process - Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of the manufacture. A minimum nitrogen content of 13.00 percent is permissable when reworked propellant is one of the basic materials of the process.

6.3 Ordering Data - See MIL-A-48078.

6.4 Submission of Inspection Equipment Designs for Approval - See MIL-A-48078. Submit equipment designs, as required to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T, Dover, New Jersey 07801.

6.5 Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the contracting officer to: Commander, ATTN: SARPA-QA-A-P, Picatinny Arsenal, Dover, N. J. 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1376-A061

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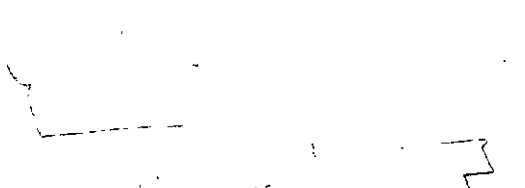
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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